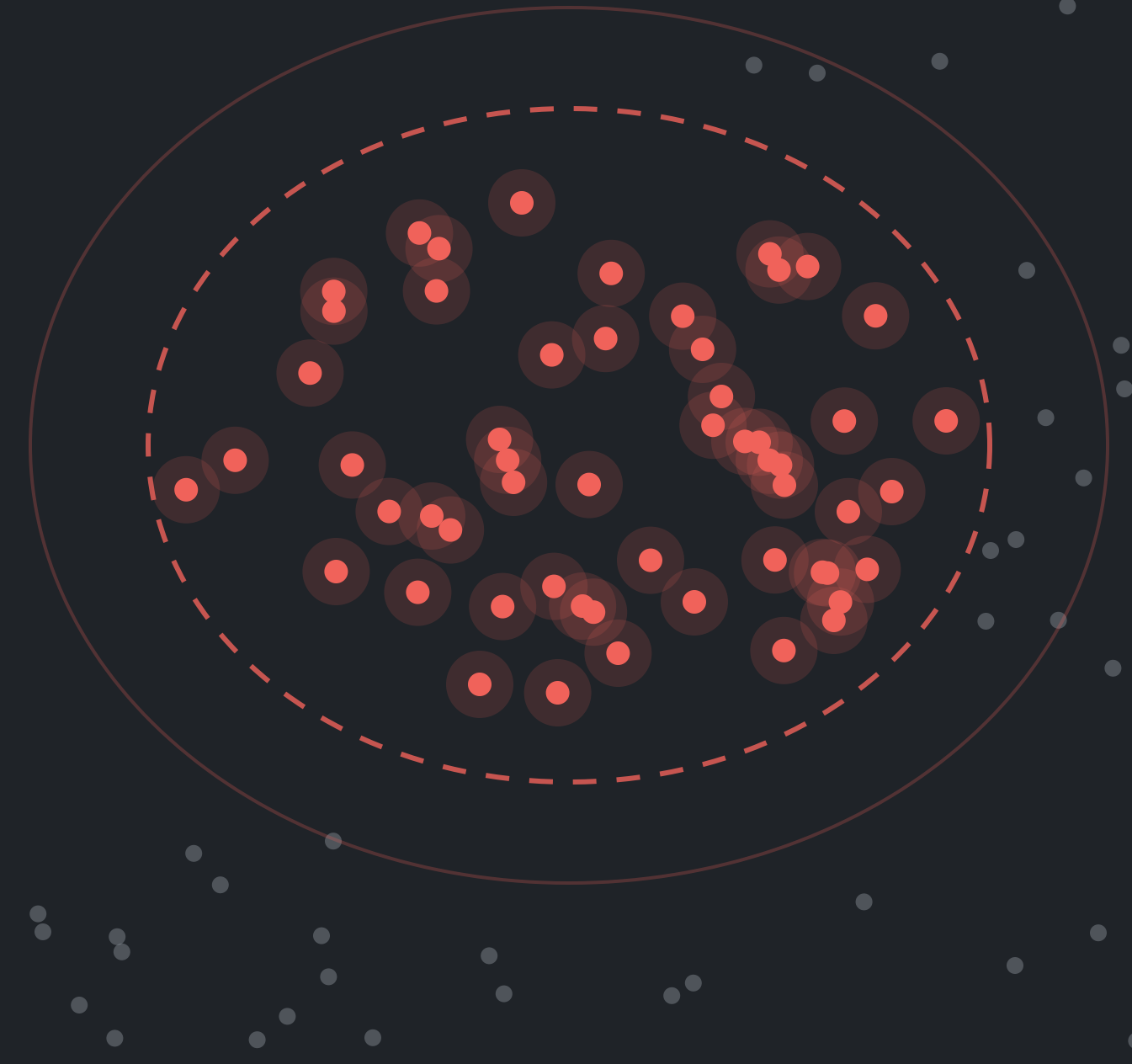


Inbox Atlas

Your agents read 4,000 tokens to find one email. Inbox Atlas hands them the 250 that matter, or tells them nothing is there.

Avinash Senthil, Antranig Baghdassarian, Chelsea Lin, Emily Han

BigRed//Hacks 2026, Navigation



Search matches words, not meaning.

🔍 coding competition

Keyword search returns

HackerRank for Stripe
Online assessment for Software Engineering
Intern **a job test**

and misses

Codeforces
Codeforces Round 1043 (Div. 2) starts in 24 hours

Cornell ACM
ICPC regional team tryouts this Saturday

Devpost
Your DevPost submission was received

Your agent does the same, only more expensively

`grep "competition"` nothing

`grep "contest"` opens 6 files

`grep "hackathon"` opens 9 more

`cat inbox/*.eml` pastes them all

And when the answer isn't there, it keeps looking.

7

of our 12 questions

Keyword search found nothing. Not wrong answers. Nothing.

Inbox Atlas found the right emails for all 12: Recall@10 **0.867** vs **0.175**, about five times the recall.

travel plans

upcoming exams

internship hunting

stuff I bought online

workout schedule

catching up with my parents

winter break trip

A question becomes a region.

You ask

typed, spoken, texted, or called by an agent

“coding competition”

→ Grok writes facets

in the words emails actually use, plus what to exclude

Codeforces round

ICPC regional

LeetCode contest

Devpost submission

hackathon

not: job coding interview

not: take-home coding test

→ Region on the map

our encoder scores every email and note against it



RELATED: YES

→ A decision

only the sentences that answer, under a token budget

Codeforces Round 1043 starts in 24 hours. ICPC tryouts this Saturday.

249 tokens

or "nothing here", so the agent stops

31 tokens

Live demo

Browser

Search by meaning

Keyword vs region on
“coding competition”, the
facet chips, then “yacht
maintenance”: NO.

iMessage

Just text it

“what do I have tomorrow?”
then “did anyone ask me for
money?”

Cursor

One MCP tool

“what pulse widths did I use
for the arm servos?” then
“did I book a yacht charter?”

18x fewer tokens than grep.

Questions with zero relevant hits

12 meaning based questions, lower is better



7 to 0

Recall@10

same 12 questions, higher is better



about 5x

Context tokens per question

top 10 docs vs 800 token pack



18x fewer

Tokens when the topic is absent

14 absent topics



60x fewer

Real 419 note vault

tokens handed to the agent



17x fewer

RAG always returns k chunks. We return a decision.

What changes	Standard RAG	Inbox Atlas	Measured
Knows when nothing is there	Always returns k chunks, even for topics that do not exist	Calibrated yes or no; abstains and says stop searching	31 vs 742 tokens on absent topics
A question is a region	One query vector; the job test ranks first	Grok facets plus exclusions push the job test out	7 vs 6 of top 8 on topic
Sentences under a budget	Pastes 8 whole chunks	Keeps only the sentences that answer	249 vs 740 tokens at the same 75% accuracy
Folders first, then notes	Flat list of chunks across the vault	Finds the folder, then the note, then the line	210 tokens to the exact line
Encoder trained for mail	Off the shelf embeddings	Frozen bge + LoRA, distilled from a frozen teacher	0.692 vs 0.635 nDCG@10, vague queries

For agents, it's one MCP tool.

Cursor

Claude Code

Hermes

OpenClaw

any second brain

"what pulse widths did I use for the arm servos?"

📁 projects/

📁 robotics/

📁 arm/

📄 **Servo calibration.md#pulse-widths**

Elbow servo: 610 to 2380 microseconds.

210 tokens

Folder first, then the note, then the exact line, in a nested markdown vault.

23 tokens

"did I book a yacht charter?" Nothing here. No grep loop. It just stops.

For people: search it, say it, text it.

Search with the map

Facets you can edit, a yes or no, and the region glowing on your inbox.

Talk or dictate

Grok realtime voice, and hold to dictate anywhere with the "um, no wait" cleaned out.

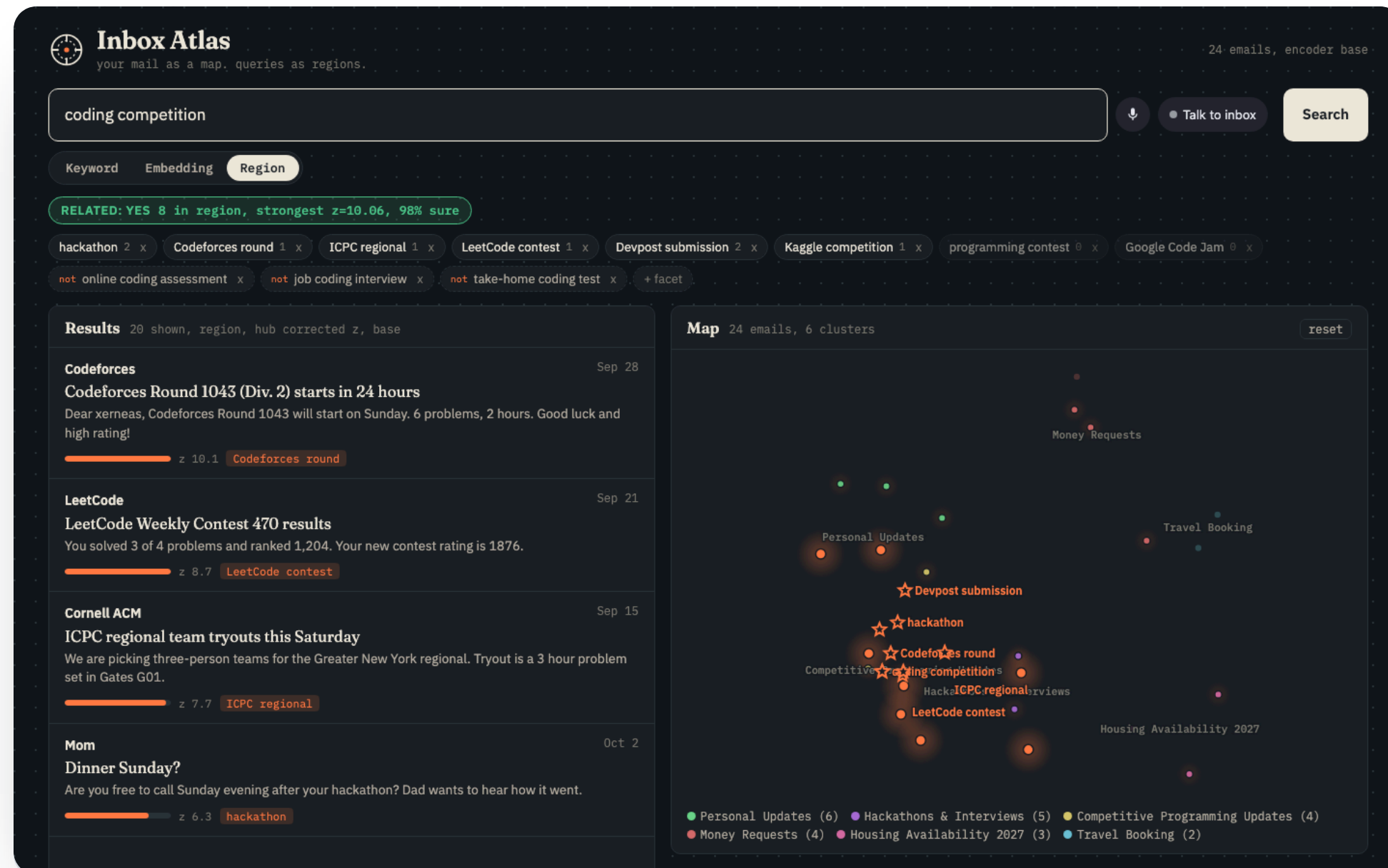
Text it on iMessage

Two-way on Photon: a morning brief, and alerts when mail lands in a topic you watch.

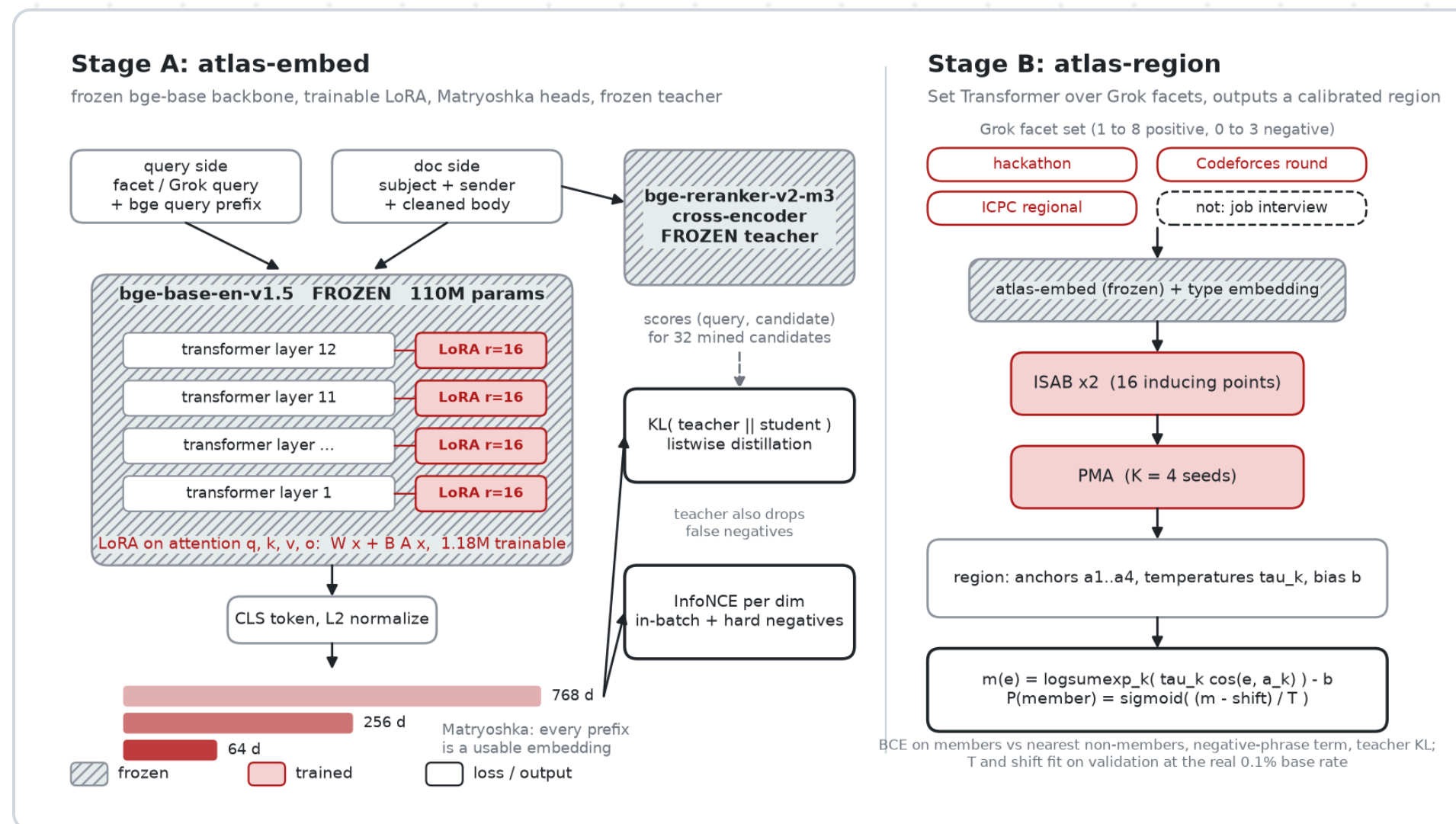
what do I have tomorrow?

Monday 10am: lab meeting with Priya.

watch internships



Our own encoder, trained overnight.



1.18M

trainable LoRA parameters
on a frozen bge-base
backbone

224,867

Enron emails

491k

teacher scores

Frozen

cross-encoder teacher,
Matryoshka heads at 768,
256, 64

19,980

Grok labels

1 GPU

RTX 4070 SUPER,
overnight

0.692 vs 0.635

nDCG@10 on vague queries, unseen emails, vs the
frozen base model

Built on

Grok

Facet writing, the tool-calling agent, speech to text and the realtime voice agent.

Tiger Cloud

Postgres with a pgvector HNSW index, a TimescaleDB hypertable logging every query, and chat history.

Photon Spectrum

The two-way iMessage agent: answers, morning brief and watch alerts.

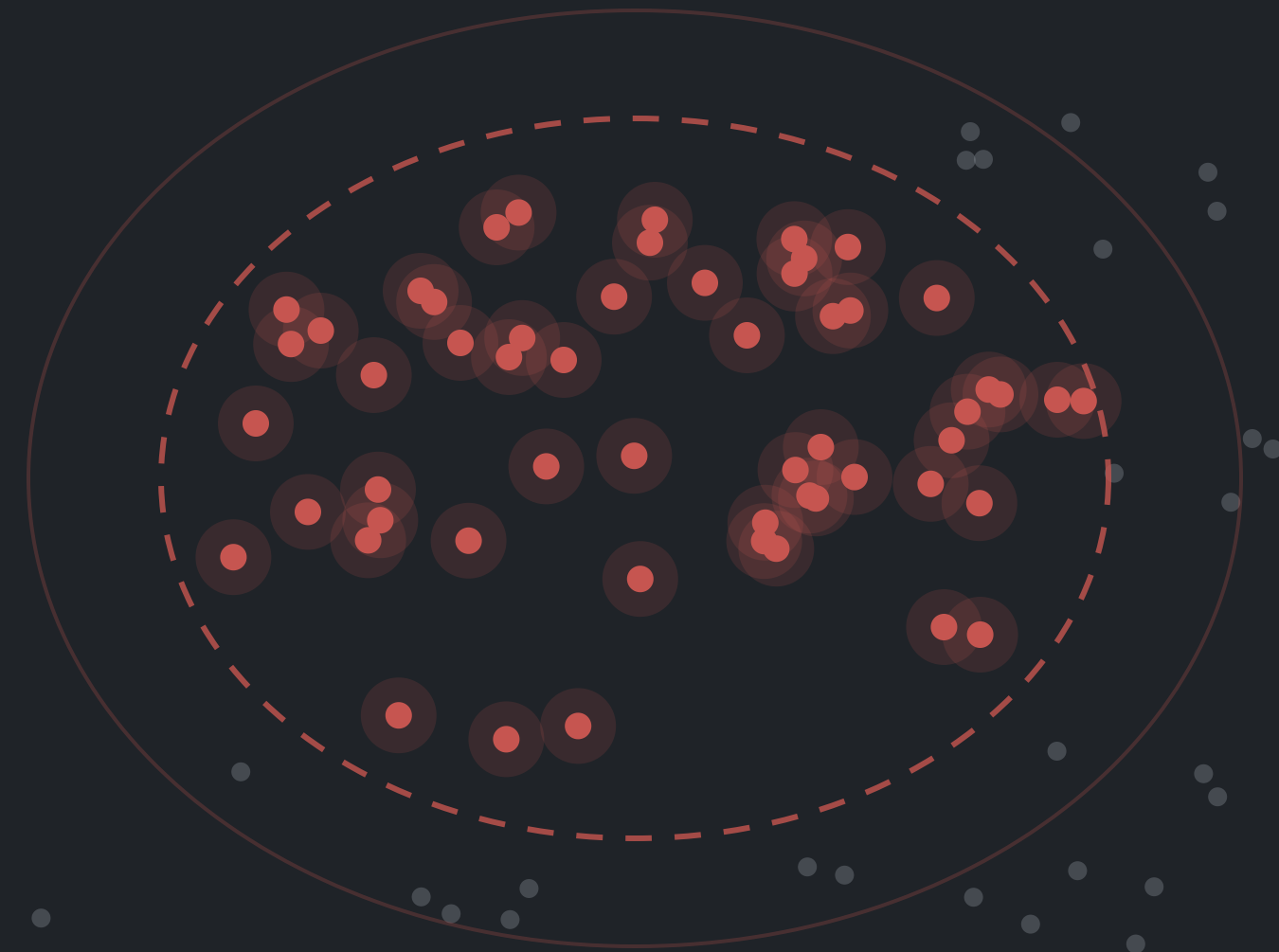
MCP

One server for Cursor, Claude Code, Hermes and OpenClaw, over stdio or HTTP.

Cursor CLI

Built and routed with it. In our own testing outside this repo, Cursor CLI was the most token-efficient at routing compared with Claude Code, Codex and Grok.

Navigation used to mean maps.
Now it also means navigating what we
and our agents know.



github.com/xerneas3318/inbox-atlas

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